



# The Ultimate SQL Master Cheat Sheet

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# 1. SQL Basics: The Foundation

Every complex data pipeline starts with mastering the foundational commands. SQL is a declarative language; your job is to tell the database *what* data you need, and the engine will figure out *how* to get it.

## 1.1 Retrieving Data (SELECT & LIMIT)

The SELECT statement is the starting point of nearly every query. It dictates exactly which columns you want to pull from a table. LIMIT (or TOP in some SQL dialects) restricts the number of rows returned, which is crucial for exploring large datasets without crashing your session.

### Example 1: Selecting specific columns

```
-- Pulls only the necessary columns, saving memory and processing time.
SELECT first_name, last_name, email
FROM employees;
```

### Example 2: Using LIMIT to sample data

```
-- Quickly inspect what the raw data looks like without pulling millions of rows.
SELECT id, transaction_date, amount
FROM sales
LIMIT 10;
```

**Quick Tip:** Avoid using SELECT \* in production code. Explicitly naming your columns makes your queries run faster and prevents pipeline breaks if the underlying table schema changes.

## 1.2 Filtering Data (WHERE)

The WHERE clause is your primary tool for reducing data volume. It filters rows based on specific conditions *before* any other grouping or math happens.

### Example 1: Filtering by exact text match

```
-- Finds all users who are currently subscribed to the platform.
SELECT user_id, email
FROM users
WHERE subscription_status = 'Active';
```

### Example 2: Filtering by numerical thresholds

```
-- Identifies high-value transactions for the fraud team.
SELECT transaction_id, amount, currency
FROM payments
WHERE amount > 5000;
```

## 1.3 Sorting Results (ORDER BY)

Data is stored in databases in an unpredictable order. If you need your results sorted chronologically or alphabetically, you must use `ORDER BY`. The default sort is ascending (`ASC`), but you can explicitly request descending order (`DESC`).

### Example 1: Sorting by a single column

```
-- Retrieves the newest products added to the catalog first.
SELECT product_name, category, created_at
FROM products
ORDER BY created_at DESC;
```

### Example 2: Multi-level sorting

```
-- Sorts first by department alphabetically, then by salary from highest to lowest.
SELECT department, employee_name, salary
FROM employees
ORDER BY department ASC, salary DESC;
```

## 1.4 Clean Outputs (Aliases & DISTINCT)

Presentation matters. **Aliases** (`AS`) allow you to rename columns on the fly for better readability. **DISTINCT** filters out duplicate rows, ensuring you only see unique values.

### Example 1: Creating readable column names with Aliases

```
-- Renames technical column names into business-friendly terms.
SELECT
    emp_id AS employee_id,
    hire_dt AS hire_date,
    base_sal AS base_salary
FROM hr_records;
```

### Example 2: Finding unique values with DISTINCT

```
-- Extracts a clean list of all countries where we have active customers.
```

```
SELECT DISTINCT country
FROM customers
WHERE is_active = true;
```

**Quick Tip:** You can also use aliases on table names (e.g., FROM users AS u). This becomes absolutely essential when you start writing JOIN clauses later on!

## 1.5 Putting It All Together: Production-Ready Examples

Here is how Senior Data Engineers combine these foundational elements into clean, efficient queries.

**Scenario A: The Marketing Pull** *The marketing team needs a list of the 50 most recent premium sign-ups from Canada to send a welcome email.*

```
SELECT DISTINCT
  user_id,
  email AS contact_email,
  signup_date
FROM users
WHERE country = 'Canada'
  AND plan_tier = 'Premium'
ORDER BY signup_date DESC
LIMIT 50;
```

**Scenario B: The Inventory Check** *The operations team needs to see the highest-priced items currently sitting in the 'Electronics' warehouse.*

```
SELECT
  sku_id AS product_code,
  product_name,
  retail_price
FROM inventory
WHERE warehouse_location = 'Electronics'
  AND stock_count > 0
ORDER BY retail_price DESC
LIMIT 100;
```

## 2. Filtering & Conditions: Precision Data Extraction

A junior analyst pulls all the data into Python or Excel to filter it. A Senior Data Engineer filters the data at the database level to save compute costs and network bandwidth. Mastering the WHERE clause logic is how you build efficient, production-grade pipelines.

### 2.1 Logical Operators (AND, OR, NOT)

When a single condition isn't enough, you need to combine them.

- **AND:** The row must meet *all* conditions.
- **OR:** The row must meet *at least one* condition.
- **NOT:** Excludes rows that match the condition.

#### Example 1: Mixing AND & OR (The Parentheses Rule)

```
-- Always use parentheses when mixing AND and OR to enforce the order of logic!
-- Without parentheses, the database might evaluate the OR before the AND.
SELECT user_id, subscription_tier, last_login
FROM users
WHERE (subscription_tier = 'Pro' OR subscription_tier = 'Enterprise')
    AND last_login >= '2026-01-01';
```

#### Example 2: Using NOT to exclude specific groups

```
-- Filters out internal test accounts from our revenue metrics.
SELECT order_id, amount, customer_id
FROM orders
WHERE NOT customer_id = 9999
    AND status = 'Completed';
```

**Quick Tip:** While `NOT column = value` works, it is much more common and readable in the industry to use the "not equal to" operators: `!=` or `<>`.

### 2.2 Streamlining Logic (IN & BETWEEN)

Writing OR statements over and over is tedious and error-prone. SQL provides clean shorthand functions for ranges and lists.

- **IN:** Checks if a value matches anything in a specific list. It is essentially a cleaner version of multiple OR statements.

- BETWEEN: Filters within a range. Crucial note: BETWEEN is *inclusive*, meaning it includes both the start and end values.

### Example 1: Filtering by a list of categories using IN

```
-- Clean, readable way to grab specific marketing segments.
SELECT campaign_id, target_audience, budget
FROM marketing_campaigns
WHERE target_audience IN ('Gen Z', 'Millennials', 'Tech Enthusiasts');
```

### Example 2: Capturing a date range using BETWEEN

```
-- Grabs all transactions for Q1.
-- Note: '2026-03-31' is included in the results.
SELECT transaction_id, amount, transaction_date
FROM sales
WHERE transaction_date BETWEEN '2026-01-01' AND '2026-03-31';
```

## 2.3 Pattern Matching (LIKE)

Sometimes you don't know the exact string, but you know the pattern. LIKE allows you to search using wildcards.

- % (Percent sign): Matches zero or more characters.
- \_ (Underscore): Matches exactly *one* character.

### Example 1: Finding domains using the % wildcard

```
-- Finds anyone who signed up with a corporate or educational email.
SELECT first_name, last_name, email
FROM leads
WHERE email LIKE '%.edu' OR email LIKE '%.gov';
```

### Example 2: Finding strict formats using the \_ wildcard

```
-- Finds invoice numbers that start with 'INV-', followed by exactly 4 characters.
-- Matches 'INV-1234', but ignores 'INV-12345'.
SELECT invoice_id, total_due
FROM invoices
WHERE invoice_code LIKE 'INV-____';
```

## 2.4 The NULL Trap (IS NULL & IS NOT NULL)

In SQL, NULL does not mean zero, and it does not mean an empty string. NULL means Unknown or Missing. Because it is unknown, you cannot use standard operators like = or != on it.

*If you write `WHERE phone_number = NULL`, SQL will return zero rows, even if there are missing phone numbers!*

### Example 1: Finding missing data for data quality checks

```
-- Identifies users who skipped the optional profile setup step.
SELECT user_id, username
FROM user_profiles
WHERE phone_number IS NULL
   OR date_of_birth IS NULL;
```

### Example 2: Ensuring data integrity before a calculation

```
-- Only calculates shipping times for orders that have actually been delivered.
SELECT order_id, dispatch_date, delivery_date
FROM shipments
WHERE delivery_date IS NOT NULL;
```

**Quick Tip:** Handling NULL values is one of the most common reasons data pipelines fail. Always ask yourself: *"What should happen if this column is NULL?"* when writing your queries.

## 2.5 Putting It All Together: Production-Ready Examples

Let's look at how these filtering concepts combine in real-world Data Engineering scenarios.

**Scenario A: The Churn Risk Analysis** *The Customer Success team needs a list of high-value customers who haven't logged in recently and haven't assigned a dedicated account manager.*

```
SELECT
  customer_id,
  company_name,
  last_login_date
FROM b2b_customers
WHERE mrr_tier IN ('Gold', 'Platinum')
   AND last_login_date BETWEEN '2025-01-01' AND '2025-12-31'
   AND account_manager_id IS NULL
   AND status = 'Active';
```

**Scenario B: The Fraud Detection Trigger** *The Security team is looking for suspicious transactions: large amounts originating outside of North America, or any transaction missing a verified IP address.*

```
SELECT
  transaction_id,
  user_id,
  amount
FROM transactions
WHERE (amount > 10000 AND country NOT IN ('USA', 'CAN'))
      OR (ip_address IS NULL)
      OR (device_id LIKE 'TEST-%');
```

## 3. Aggregations: Turning Raw Rows into Business Insights

Raw data is rarely useful to stakeholders. The CEO doesn't want to see 10 million individual transaction rows; they want to see the total revenue for the quarter. Aggregations are how Data Engineers and Analysts summarize massive datasets into actionable metrics.

### 3.1 The "Big Five" Aggregate Functions

SQL provides five core functions to perform mathematical operations across multiple rows. When used without a `GROUP BY` clause, these functions condense your entire table into a single row of metrics.

- **COUNT()**: Counts the number of rows.
- **SUM()**: Adds up all the numeric values in a column.
- **AVG()**: Calculates the mean average of a numeric column.
- **MIN()**: Finds the lowest value (works on numbers, dates, and text).
- **MAX()**: Finds the highest value (works on numbers, dates, and text).

#### Example 1: Building a high-level summary dashboard

```
-- Generates a single-row snapshot of the company's entire sales history.
SELECT
  COUNT(order_id) AS total_orders,
  SUM(amount) AS total_revenue,
  AVG(amount) AS average_order_value,
  MIN(order_date) AS first_sale_date,
  MAX(order_date) AS latest_sale_date
FROM sales;
```

#### Example 2: The COUNT(\*) vs COUNT(column) distinction

```
-- COUNT(*) counts EVERY row.
-- COUNT(phone_number) ONLY counts rows where phone_number is NOT NULL.
-- This is a very common Data Engineer interview question!
SELECT
  COUNT(*) AS total_registered_users,
  COUNT(phone_number) AS users_with_valid_phones
FROM users;
```

**Quick Tip:** To count the number of *unique* values in a column, combine `COUNT` with `DISTINCT`. For example: `SELECT COUNT(DISTINCT country) FROM users;`

## 3.2 Slicing the Data (GROUP BY)

While global summaries are great, business questions usually require slicing the data into categories. The GROUP BY clause bundles rows that have the same values into summary rows.

Rule of Thumb: If you have an aggregate function (like SUM()) and a regular column (like department) in your SELECT statement, that regular column must be in the GROUP BY clause.

### Example 1: Grouping by a single dimension

```
-- Calculates the total revenue generated by each specific product category.
SELECT category, SUM(price) AS total_sales
FROM products
GROUP BY category;
```

### Example 2: Grouping by multiple dimensions (Drill-down)

```
-- Breaks down sales not just by region, but by the year the sale occurred.
SELECT
  region,
  EXTRACT(YEAR FROM sale_date) AS sale_year,
  COUNT(order_id) AS total_transactions
FROM regional_sales
GROUP BY region, EXTRACT(YEAR FROM sale_date)
ORDER BY region ASC, sale_year DESC;
```

## 3.3 Filtering Aggregations (HAVING)

One of the most common beginner mistakes is trying to use a WHERE clause to filter the result of a SUM() or COUNT().

The database processes the WHERE clause before it calculates aggregations. Therefore, if you want to filter your data after it has been grouped and summarized, you must use the HAVING clause.

- WHERE: Filters individual rows going into the aggregate.
- HAVING: Filters the groups coming out of the aggregate.

### Example 1: Using HAVING to find outliers

```
-- We want to find IP addresses that have pinged our server more than 500 times.
-- We cannot use WHERE request_count > 500 because the count hasn't happened yet!
SELECT ip_address, COUNT(*) AS request_count
FROM server_logs
```

```
GROUP BY ip_address  
HAVING COUNT(*) > 500;
```

### Example 2: Combining WHERE and HAVING

```
-- Filters for active users (WHERE), calculates their total spend,  
-- and then only shows users who spent more than $1,000 (HAVING).  
SELECT user_id, SUM(purchase_amount) AS lifetime_value  
FROM transactions  
WHERE account_status = 'Active'  
GROUP BY user_id  
HAVING SUM(purchase_amount) > 1000;
```

## 3.4 Putting It All Together: Production-Ready Examples

Let's see how a Data Engineer writes robust aggregation queries for downstream reporting tools like Tableau or Looker.

**Scenario A:** The E-Commerce Cohort Analysis The product team needs a report showing the average rating and total reviews for items in the 'Electronics' and 'Home' categories, but they only care about products that have received at least 50 reviews.

```
SELECT  
  product_category,  
  product_name,  
  COUNT(review_id) AS total_reviews,  
  AVG(star_rating) AS average_rating  
FROM product_reviews  
WHERE product_category IN ('Electronics', 'Home')  
GROUP BY product_category, product_name  
HAVING COUNT(review_id) >= 50  
ORDER BY average_rating DESC;
```

**Scenario B:** The HR Salary Audit The finance department is auditing payroll. They need to see the minimum, maximum, and average salary for each department, but only for full-time employees. They also want to exclude any department where the average salary is below \$60,000.

```
SELECT  
  department_name,  
  COUNT(employee_id) AS headcount,  
  MIN(base_salary) AS lowest_salary,  
  MAX(base_salary) AS highest_salary,  
  AVG(base_salary) AS average_salary
```



```
FROM payroll_records  
WHERE employment_type = 'Full-Time'  
GROUP BY department_name  
HAVING AVG(base_salary) >= 60000  
ORDER BY average_salary DESC;
```

## 4. Joins: The Heart of Relational Data (Very Important)

In the real world, data is "normalized"—split into multiple tables to save space and maintain accuracy. As a Data Engineer, your primary job is stitching this data back together using JOIN clauses. Think of a Join as a horizontal combination of tables based on a shared key.

### The Setup: Our Sample Tables

To understand joins, visualize these two simple tables. We will use them for the next few examples.

Table A: employees

| emp_id | emp_name | dept_id |
|--------|----------|---------|
| 1      | Alice    | 10      |
| 2      | Bob      | 20      |
| 3      | Charlie  | NULL    |

(Charlie hasn't been assigned a dept)

Table B: departments

| dept_id | dept_name   |
|---------|-------------|
| 10      | HR          |
| 20      | Engineering |
| 30      | Marketing   |

(No employees are in Marketing yet)

### 4.1 INNER JOIN (The Strict Match)

An INNER JOIN returns only the rows where there is a match in both tables.

- Visual Diagram (Venn): The overlapping middle section of two circles.
- Result from our tables: Only Alice and Bob are returned. Charlie and Marketing are dropped.

#### Example 1: Standard Inner Join

-- Retrieves employees who belong to a valid department.

```
SELECT e.emp_name, d.dept_name  
FROM employees AS e
```

```
INNER JOIN departments AS d
  ON e.dept_id = d.dept_id;
```

## 4.2 LEFT JOIN (The Data Engineer's Best Friend)

A LEFT JOIN returns all rows from the "Left" table (the one written first), and any matching rows from the "Right" table. If there is no match, the Right table's columns will show as NULL.

- Visual Diagram (Venn): The entire left circle, plus the overlapping middle.
- Result: Alice, Bob, and Charlie (with NULL for his department name).

### Example 1: Keeping the primary dataset intact

```
-- Retrieves ALL employees, showing their department if they have one.
SELECT e.emp_name, d.dept_name
FROM employees AS e
LEFT JOIN departments AS d
  ON e.dept_id = d.dept_id;
```

**Quick Tip:** LEFT JOIN is the most common join in Data Engineering. We use it to append dimensional data (like product names) to fact tables (like sales) without accidentally dropping sales records that have missing product IDs.

## 4.3 RIGHT JOIN (The Flipped Left Join)

A RIGHT JOIN is the exact opposite of a LEFT JOIN. It returns all rows from the Right table, and matched rows from the Left.

- Visual Diagram (Venn): The entire right circle, plus the overlapping middle.
- Result: HR, Engineering, and Marketing (with NULL for the employee name).

### Example 1: Finding orphaned records

```
-- Retrieves ALL departments, even if no employees work there.
SELECT e.emp_name, d.dept_name
FROM employees AS e
RIGHT JOIN departments AS d
  ON e.dept_id = d.dept_id;
```

**(Note:** Most professionals just use LEFT JOIN and flip the order of the tables for readability, making RIGHT JOIN quite rare in production code).

## 4.4 FULL OUTER JOIN (The Complete Picture)

A FULL OUTER JOIN returns all rows from both tables. If there's a match, they are joined. If not, the missing side is filled with NULLs.

- Visual Diagram (Venn): Both entire circles, completely filled in.
- Result: Alice, Bob, Charlie (NULL dept), and Marketing (NULL employee).

### Example 1: The comprehensive audit

```
-- Useful for auditing disparate systems to find missing links on both sides.  
SELECT e.emp_name, d.dept_name  
FROM employees AS e  
FULL OUTER JOIN departments AS d  
  ON e.dept_id = d.dept_id;
```

## 4.5 SELF JOIN (The Hierarchy Solver)

A SELF JOIN isn't a new keyword; it's the technique of joining a table to itself. You must use aliases to trick SQL into treating it like two separate tables.

- Use Case: Hierarchical data (e.g., matching employees to their managers, or categories to sub-categories).

### Example 1: The Manager-Employee Lookup

```
-- Assuming the employees table has a 'manager_id' column.  
-- We join the table to itself to look up the manager's name.  
SELECT  
  worker.emp_name AS employee,  
  boss.emp_name AS manager  
FROM employees AS worker  
LEFT JOIN employees AS boss  
  ON worker.manager_id = boss.emp_id;
```

## 4.6 Putting It All Together: Production-Ready Examples

In the real world, you rarely join just two tables. You often chain multiple joins together to build a complete "wide" dataset for reporting.

**Scenario A:** The E-Commerce Master View The analytics team needs a report showing order details, customer emails, and the name of the product purchased. We must ensure we don't lose any orders, even if the product was deleted from the system (hence the LEFT JOINS).

```
SELECT
  o.order_id,
  o.order_date,
  c.email AS customer_email,
  p.product_name,
  o.total_price
FROM orders AS o
INNER JOIN customers AS c
  ON o.customer_id = c.customer_id
LEFT JOIN products AS p
  ON o.product_id = p.product_id
WHERE o.order_date >= '2026-01-01';
```

**Scenario B:** The "Anti-Join" (Finding Missing Data) Marketing wants to send a promo code to users who created an account but have never placed an order. We use a LEFT JOIN and filter for NULLs.

```
SELECT
  u.user_id,
  u.email
FROM users AS u
LEFT JOIN orders AS o
  ON u.user_id = o.user_id
WHERE o.order_id IS NULL; -- This guarantees they have no matching orders
```

## 5. Subqueries & CTEs: Mastering Nested Logic

Sometimes, a single query isn't enough. You often need to perform an aggregation or filter before you can run your main query. Both Subqueries and Common Table Expressions (CTEs) allow you to query the results of another query, but they handle readability and execution very differently.

### 5.1 Inline Subqueries (The Traditional Method)

A subquery is simply a query nested inside another query. They are typically enclosed in parentheses and can be placed in the SELECT, FROM, or WHERE clauses.

While powerful, subqueries can quickly become difficult to read—often referred to as the "onion problem" because you have to read them from the inside out.

#### Example 1: Subquery in the WHERE clause

```
-- Finds all employees who earn more than the company-wide average salary.
-- The database calculates the inner query FIRST, then uses that scalar value for the outer query.
SELECT employee_name, department, salary
FROM employees
WHERE salary > (
    SELECT AVG(salary)
    FROM employees
);
```

#### Example 2: Subquery in the FROM clause (Derived Tables)

```
-- Calculates the average revenue across different regions.
-- We must aggregate the total revenue per region first, then average those totals.
SELECT AVG(regional_total) AS avg_regional_revenue
FROM (
    SELECT region, SUM(revenue) AS regional_total
    FROM sales
    GROUP BY region
) AS regional_sales_summary; -- Every subquery in a FROM clause MUST have an alias!
```

**Quick Tip:** Be careful with Correlated Subqueries (a subquery that references a column from the outer query). They force the database to re-run the subquery for every single row of the outer query, which can devastate performance on large datasets.

## 5.2 Common Table Expressions / CTEs (The Senior Standard)

A Common Table Expression (CTE) uses the WITH clause to create a temporary, named result set that exists only for the duration of the query.

In modern Data Engineering, CTEs are strictly preferred over nested subqueries. They allow you to read code top-to-bottom, just like a Python script, making complex logic highly readable and debuggable.

### Example 1: The WITH Clause syntax

-- This does the exact same thing as Example 2 above, but is much easier to read.

```
WITH regional_sales_summary AS (  
  SELECT region, SUM(revenue) AS regional_total  
  FROM sales  
  GROUP BY region  
)  
SELECT AVG(regional_total) AS avg_regional_revenue  
FROM regional_sales_summary;
```

### Example 2: Chaining Multiple CTEs

-- You can define multiple CTEs by separating them with a comma.

-- You DO NOT write the word 'WITH' a second time.

```
WITH high_value_customers AS (  
  SELECT customer_id, SUM(amount) AS total_spent  
  FROM orders  
  GROUP BY customer_id  
  HAVING SUM(amount) > 10000  
,  
recent_activity AS (  
  SELECT customer_id, MAX(login_date) AS last_login  
  FROM sessions  
  GROUP BY customer_id  
)  
-- The final query stitches the CTEs together  
SELECT hvc.customer_id, hvc.total_spent, ra.last_login  
FROM high_value_customers AS hvc  
INNER JOIN recent_activity AS ra  
  ON hvc.customer_id = ra.customer_id;
```

## 5.3 Real-World Use Cases & Best Practices

When should you use a CTE? Almost always, especially when building reporting pipelines, handling multi-step transformations, or when you need to reference the same temporary dataset more than once in a single query.

**Scenario A:** The Pre-Aggregation Join (Performance Boost) If you join a massive 1-billion-row sales table to a user table, the database has to join all 1 billion rows. A Senior Data Engineer uses a CTE to aggregate the sales data first, reducing it to a few thousand rows BEFORE the join.

```
WITH aggregated_sales AS (
  -- Step 1: Crush 1 billion rows down to 1 row per user
  SELECT user_id, COUNT(order_id) AS total_orders, SUM(amount) AS lifetime_value
  FROM transactions
  WHERE status = 'Completed'
  GROUP BY user_id
)
-- Step 2: Join the tiny, aggregated dataset to the users table
SELECT u.email, s.total_orders, s.lifetime_value
FROM users u
INNER JOIN aggregated_sales s
  ON u.id = s.user_id;
```

**Scenario B:** Deduplicating Data (The Row Number Trick) We have duplicate transaction logs and need to keep only the most recent entry for each transaction ID. We use a CTE combined with a window function (covered in the next section) to identify and filter the dupes.

```
WITH ranked_transactions AS (
  SELECT
    transaction_id,
    status,
    updated_at,
    ROW_NUMBER() OVER(PARTITION BY transaction_id ORDER BY updated_at DESC)
  AS rn
  FROM raw_transaction_logs
)
-- Only select the rows where the rank is 1 (the most recent update)
SELECT transaction_id, status, updated_at
FROM ranked_transactions
WHERE rn = 1;
```



**Pro Tip:** Are CTEs faster than subqueries? In modern database engines (like Snowflake, BigQuery, or PostgreSQL), the query optimizer usually treats them exactly the same under the hood. You use CTEs for human readability and maintainability, not necessarily for execution speed.

## 6. Window Functions: The Senior Engineer's Secret Weapon

Standard aggregations (using `GROUP BY`) collapse your data—if you have 100 rows and group them by 5 categories, your output is only 5 rows.

**Window Functions** are different. They allow you to perform calculations across a set of related rows (a "window") *without* collapsing the individual rows. You get to keep your row-level detail while adding advanced analytical columns right next to it.

### 6.1 The Anatomy of a Window Function (`OVER & PARTITION BY`)

Every window function requires the `OVER()` clause. Inside the `OVER()` clause, you define exactly how the "window" of data should be framed using two optional keywords:

- **PARTITION BY**: Divides the result set into separate groups (similar to `GROUP BY`). The calculation restarts for every new partition.
- **ORDER BY**: Determines the logical order of the rows *inside* that specific partition.

#### Example 1: The standard syntax

-- Calculates the average salary for the department, but keeps the individual employee rows intact.

```
SELECT
  employee_name,
  department,
  salary,
  AVG(salary) OVER(PARTITION BY department) AS dept_avg_salary
FROM employees;
```

### 6.2 Numbering and Ranking (`ROW_NUMBER`, `RANK`, `DENSE_RANK`)

These three functions are used to assign a ranking integer to rows based on the `ORDER BY` clause within your window. They handle "ties" (identical values) very differently.

- **ROW\_NUMBER()**: Assigns a strictly unique, sequential integer (1, 2, 3, 4). If there is a tie, it arbitrarily picks which gets which number.
- **RANK()**: Gives tied values the same ranking, but *skips* the next numbers (1, 2, 2, 4).
- **DENSE\_RANK()**: Gives tied values the same ranking, and *does not skip* numbers (1, 2, 2, 3).

### Example 1: Seeing the difference side-by-side

-- Assume Alice and Bob both make \$90,000, and Charlie makes \$80,000.

```
SELECT
  employee_name,
  salary,
  ROW_NUMBER() OVER(ORDER BY salary DESC) AS row_num,    -- Result: 1, 2, 3
  RANK() OVER(ORDER BY salary DESC) AS standard_rank,    -- Result: 1, 1, 3 (skips 2)
  DENSE_RANK() OVER(ORDER BY salary DESC) AS dense_rank  -- Result: 1, 1, 2 (no
gaps)
FROM employees;
```

## 6.3 Cumulative Metrics (Running Totals)

In financial reporting and inventory management, you rarely just look at daily numbers; you need cumulative totals over time. Window functions make this trivial by using an `ORDER BY` clause *without* a `PARTITION BY`.

### Example 1: Calculating a running revenue total

-- As the date increases, the `cumulative_revenue` column adds the current day to all previous days.

```
SELECT
  transaction_date,
  daily_revenue,
  SUM(daily_revenue) OVER(ORDER BY transaction_date ASC) AS cumulative_revenue
FROM daily_sales;
```

### Example 2: Running totals reset by month

-- By adding `PARTITION BY month`, the running total drops back to \$0 at the start of every new month.

```
SELECT
  transaction_date,
  sales_month,
  daily_revenue,
  SUM(daily_revenue) OVER(PARTITION BY sales_month ORDER BY transaction_date ASC)
AS month_to_date_revenue
FROM daily_sales;
```

## 6.4 Real-World Data Engineering Use Cases

In production pipelines, window functions are used daily for data cleaning and complex reporting. Here is how professionals use them.

**Scenario A: The "Top N" per Category (Great Interview Question)** *The marketing team wants to reward the top 3 highest-spending customers in every single country. We use a CTE combined with DENSE\_RANK.*

```
WITH RankedCustomers AS (
  SELECT
    customer_id,
    country,
    total_spent,
    DENSE_RANK() OVER(PARTITION BY country ORDER BY total_spent DESC) AS
    spend_rank
  FROM customer_lifetime_value
)
-- We must filter in the outer query because window functions cannot be used in a WHERE
-- clause!
SELECT customer_id, country, total_spent
FROM RankedCustomers
WHERE spend_rank <= 3;
```

**Scenario B: Data Deduplication (Keeping the Latest Record)** *In modern data lakes, updates are often appended as new rows. We have 5 rows for user\_id = 123 because they updated their profile 4 times. We only want the most recent row.*

```
WITH DeduplicatedUsers AS (
  SELECT
    user_id,
    email,
    profile_status,
    updated_at,
    ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY updated_at DESC) AS rn
  FROM raw_user_updates
)
-- By grabbing rn = 1, we guarantee exactly one row per user (the most recent one).
SELECT user_id, email, profile_status
FROM DeduplicatedUsers
WHERE rn = 1;
```



**Pro Tip:** Deduplication using `ROW_NUMBER() OVER(PARTITION BY id ORDER BY timestamp DESC)` is arguably the single most written query in a Data Engineer's day-to-day job. Memorize this pattern!

## 7. Data Modification (DML): Changing the Database

Up to this point, we have used `SELECT` to read data without altering the underlying database. Now we move into **Data Manipulation Language (DML)**. When using `INSERT`, `UPDATE`, and `DELETE`, you are actively changing production data. *With great power comes great responsibility.*

### 7.1 Adding New Rows (`INSERT INTO`)

The `INSERT` statement appends new records to a table. It is best practice to explicitly declare the columns you are inserting into. If the schema changes in the future (e.g., a new column is added), omitting the column names will break your data pipeline.

#### Example 1: The standard single-row and multi-row insert

```
-- Inserting a single record.
INSERT INTO customers (first_name, last_name, email, signup_date)
VALUES ('John', 'Doe', 'john.doe@email.com', '2026-03-28');

-- You can chain multiple records together to save execution time.
INSERT INTO customers (first_name, last_name, email, signup_date)
VALUES
  ('Jane', 'Smith', 'jane.s@email.com', '2026-03-28'),
  ('Alice', 'Wonder', 'alice.w@email.com', '2026-03-28');
```

#### Example 2: The Data Engineer's Insert (`INSERT INTO ... SELECT`)

```
-- Instead of typing out values manually, DEs constantly move data from one table to another.
-- Here, we back up old inactive users into an archive table.
INSERT INTO archived_users (user_id, email, last_login)
SELECT id, email_address, last_login_date
FROM active_users
WHERE status = 'Inactive'
AND last_login_date < '2025-01-01';
```

### 7.2 Modifying Existing Rows (`UPDATE`)

The `UPDATE` statement alters existing data. **Rule #1 of SQL: Never write an `UPDATE` statement without a `WHERE` clause.** If you omit it, you will update every single row in the table.

### Example 1: Updating a single value based on a condition

```
-- A customer upgrades their subscription plan.  
UPDATE user_subscriptions  
SET plan_tier = 'Premium',  
    updated_at = CURRENT_TIMESTAMP  
WHERE user_id = 10455;
```

### Example 2: Bulk updates using mathematical operations

```
-- Applying a 10% price increase to all products in a specific category.  
UPDATE products  
SET base_price = base_price * 1.10  
WHERE category = 'Electronics'  
    AND in_stock = true;
```

## 7.3 Removing Rows (DELETE)

The DELETE statement permanently removes rows from a table. Just like UPDATE, forgetting the WHERE clause here is a resume-generating event.

### Example 1: Purging specific records

```
-- Deleting shopping cart items that have been abandoned for over 30 days.  
DELETE FROM shopping_carts  
WHERE status = 'Abandoned'  
    AND last_modified < CURRENT_DATE - INTERVAL '30 days';
```

**Quick Tip (DELETE vs. TRUNCATE):** If you need to empty an *entire* table, do not use DELETE FROM table;. Use TRUNCATE TABLE table;. TRUNCATE is a Data Definition (DDL) command that drops and recreates the table structure instantly, whereas DELETE painstakingly logs the removal of every single row, which can take hours on a billion-row table.

## 7.4 The Senior Data Engineer's Best Practices

Writing the syntax for DML is easy. Executing it safely in a production environment requires discipline.

## Best Practice #1: The "Select Before You Modify" Rule

Before you highlight and run an UPDATE or DELETE statement, write a SELECT statement using the *exact same* WHERE clause.

-- 1. Run this first to visually verify you are targeting the right 5 rows.

```
SELECT * FROM employees WHERE department_id = 99;
```

-- 2. Only after verifying the output, run the actual delete.

```
DELETE FROM employees WHERE department_id = 99;
```

## Best Practice #2: Soft Deletes over Hard Deletes

In modern applications, we rarely DELETE user data from the database. It ruins historical reporting and makes accidental deletions unrecoverable. Instead, we use a "Soft Delete" by updating a boolean flag.

-- ❌ BAD: Hard Delete (Data is gone forever)

```
DELETE FROM users WHERE id = 500;
```

-- ✅ GOOD: Soft Delete (Data remains, but is flagged as inactive)

```
UPDATE users
```

```
SET is_deleted = true,
```

```
    deleted_at = CURRENT_TIMESTAMP
```

```
WHERE id = 500;
```

*(Note: If you use soft deletes, you must remember to add WHERE is\_deleted = false to all your downstream reporting queries!)*

## Best Practice #3: Use Transactions

If you are running a script that updates multiple tables, you want them to either *all* succeed or *all* fail. If a script errors halfway through, you are left with corrupted, partial data. Wrap your DML in a transaction block.

```
BEGIN; -- Starts the transaction
```

```
UPDATE accounts SET balance = balance - 500 WHERE account_id = 1;
```

```
UPDATE accounts SET balance = balance + 500 WHERE account_id = 2;
```

```
COMMIT; -- Saves the changes permanently.
```

-- (If something went wrong, you would type ROLLBACK; to undo everything since BEGIN).

## 8. Performance Tips: Think Like a Database Engine

A junior analyst writes queries to get the correct answer. A Senior Data Engineer writes queries to get the correct answer while using the least amount of CPU, memory, and network I/O possible.

### 8.1 The SELECT \* Tax

Using SELECT \* is fine for quick ad-hoc exploration, but it is a massive anti-pattern in production pipelines.

1. **Network & I/O Waste:** If a table has 50 columns and you only need 3, pulling all 50 wastes massive amounts of memory and network bandwidth.
2. **Columnar Databases:** In modern data warehouses (like Redshift, BigQuery, or Parquet files), data is stored in columns, not rows. SELECT \* forces the engine to scan the entire storage drive instead of just the specific columns you need.
3. **Pipeline Fragility:** If someone adds a new column to the table later, SELECT \* will pull it, potentially breaking downstream applications that expect a strict schema.

#### Example 1: The Lazy Approach (✗ BAD)

-- Pulls every column, including massive text blobs or JSON columns you don't even need.

```
SELECT *  
FROM user_activity_logs  
WHERE action = 'login';
```

#### Example 2: The Production Standard (✓ GOOD)

-- Explicitly names only the required columns. Fast, efficient, and future-proof.

```
SELECT session_id, user_id, timestamp  
FROM user_activity_logs  
WHERE action = 'login';
```

### 8.2 Index Usage Basics

Think of an index like the index at the back of a textbook. If you want to find every page that mentions "SQL," you don't read the entire book cover-to-cover (a "Full Table Scan"). You check the index, find the exact page numbers, and jump straight there (an "Index Seek").

Indexes are primarily used to speed up WHERE clauses and JOIN conditions.

### Example 1: Creating an Index on frequently queried columns

-- If we frequently search for users by email, an index prevents full table scans.

```
CREATE INDEX idx_users_email  
ON users (email);
```

-- If we frequently filter by both department and status, a composite index is best.

```
CREATE INDEX idx_emp_dept_status  
ON employees (department_id, is_active);
```

### Example 2: Leveraging the Index in your queries

-- Because 'email' is indexed, the database engine finds this row instantly.

```
SELECT id, first_name, last_name  
FROM users  
WHERE email = 'ceo@smartdatacamp.com';
```

**Pro Tip:** Don't put an index on *every* column. While indexes drastically speed up SELECT queries, they slow down INSERT, UPDATE, and DELETE commands, because the database has to update the index every time the underlying data changes!

## 8.3 SARGable Queries (The #1 Optimization Trick)

**SARGable** stands for *Search Argument Able*. A query is SARGable if the database can utilize an index to execute it.

The golden rule of SARGability: **Never wrap an indexed column in a function inside your WHERE clause.** If you apply a function to a column, the database engine is forced to evaluate that function on every single row in the table *before* it can check the index, completely ruining your performance.

### Example 1: Filtering Dates (❌ BAD - Non-SARGable)

-- By using the YEAR() function on the indexed 'created\_at' column,  
-- the database must do a Full Table Scan. The index is rendered useless!

```
SELECT order_id, total  
FROM orders  
WHERE YEAR(created_at) = 2026;
```

### Example 2: Filtering Dates (✅ GOOD - SARGable)

-- By leaving the 'created\_at' column bare, the database can use the index  
-- to instantly jump to January 1st and stop at December 31st.

```
SELECT order_id, total
FROM orders
WHERE created_at >= '2026-01-01' AND created_at < '2027-01-01';
```

### Example 3: Wildcards (❌ BAD vs ✅ GOOD)

-- ❌ BAD: Wildcard at the start prevents index usage (Index Scan).  
 SELECT name FROM customers WHERE name LIKE '%Smith';

-- ✅ GOOD: Wildcard at the end allows the engine to use the index (Index Seek).  
 SELECT name FROM customers WHERE name LIKE 'Smith%';

## 8.4 Query Optimization: Filter Early, Join Late

Data Engineering is all about managing dataset sizes in memory. If you are joining massive tables together, you should aggressively filter the data *before* the join occurs.

### Example 1: The Expensive Join (❌ BAD)

-- Joins the entire 50-million-row sales table to the users table,  
 -- and ONLY THEN filters down to a single day. This consumes massive memory.  
 SELECT u.email, s.amount  
 FROM users u  
 JOIN sales s ON u.id = s.user\_id  
 WHERE s.sale\_date = '2026-03-28';

### Example 2: The Optimized Pre-Filter (✅ GOOD)

-- Uses a CTE to filter the sales table down to just a few thousand rows FIRST,  
 -- making the subsequent join to the users table lightning fast.  
 WITH todays\_sales AS (  
 SELECT user\_id, amount  
 FROM sales  
 WHERE sale\_date = '2026-03-28'  
 )  
 SELECT u.email, ts.amount  
 FROM users u  
 JOIN todays\_sales ts ON u.id = ts.user\_id;

**Quick Tip:** Before running a complex query, put the word EXPLAIN or EXPLAIN ANALYZE (depending on your SQL dialect) in front of your SELECT statement. The database won't run the



query; instead, it will print out the "Execution Plan," showing you exactly where the bottlenecks and full table scans are hiding!

## 9. Common SQL Interview Questions: The Top 20 Practical Tests

These 20 questions represent the core patterns you will face in almost any technical screening for a Data Analyst or Data Engineer role. Memorize the logic, not just the syntax.

### The Classics: Ranking & Duplicates

**Q1. How do you find the 2nd highest salary in an employees table?** Use `DENSE_RANK()` to account for ties, or use a subquery/offset for a quick pull.

```
SELECT MAX(salary) AS second_highest
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

**Q2. Write a query to find all duplicate email addresses in a table.** Group by the column in question and use `HAVING` to find counts greater than 1.

```
SELECT email, COUNT(*) AS occurrences
FROM users
GROUP BY email
HAVING COUNT(*) > 1;
```

**Q3. How do you delete duplicate rows, keeping only the one with the lowest ID?** Use a CTE with `ROW_NUMBER()` partitioned by the duplicate column.

```
WITH RankedDuplicates AS (
  SELECT id, ROW_NUMBER() OVER(PARTITION BY email ORDER BY id ASC) as rn
  FROM users
)
DELETE FROM users WHERE id IN (SELECT id FROM RankedDuplicates WHERE rn > 1);
```

**Q4. Explain the difference between `RANK()`, `DENSE_RANK()`, and `ROW_NUMBER()`.**

- `ROW_NUMBER()`: Unique sequential integers (1, 2, 3, 4).
- `RANK()`: Allows ties, but skips subsequent numbers (1, 2, 2, 4).
- `DENSE_RANK()`: Allows ties, and does *not* skip numbers (1, 2, 2, 3).

## Filtering & Aggregation Logic

**Q5. What is the difference between WHERE and HAVING?** WHERE filters individual rows *before* aggregation. HAVING filters the grouped results *after* aggregation.

**Q6. Find the top 3 highest-paid employees in *each* department.** This requires a window function. You cannot use a simple LIMIT 3 because you need 3 per group.

```
WITH RankedSalaries AS (
  SELECT name, department, salary,
         DENSE_RANK() OVER(PARTITION BY department ORDER BY salary DESC) as rank
  FROM employees
)
SELECT name, department, salary FROM RankedSalaries WHERE rank <= 3;
```

**Q7. How do you handle NULL values when doing mathematical operations?** Use the COALESCE ( ) function to convert NULL into a usable default value (like 0) so the math doesn't break.

```
SELECT employee_id, (base_salary + COALESCE(bonus, 0)) AS total_compensation
FROM payroll;
```

**Q8. Explain UNION vs. UNION ALL.** Both combine the results of two SELECT statements vertically. UNION removes exact duplicate rows (which requires a slow sorting operation). UNION ALL keeps duplicates and is significantly faster.

## Joins & Relational Data

**Q9. Write a query to find customers who have *never* placed an order.** Use a LEFT JOIN from customers to orders, and filter for rows where the order ID is NULL.

```
SELECT c.customer_id, c.name
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

**Q10. Write a query to find employees who earn more than their direct managers.** This requires a SELF JOIN, linking the employee's manager\_id to the manager's employee\_id.

```
SELECT e.name AS employee_name
FROM employees e
JOIN employees m ON e.manager_id = m.employee_id
WHERE e.salary > m.salary;
```

**Q11. What is a Cross Join (Cartesian Product)?** A CROSS JOIN joins every row in Table A to every row in Table B. If Table A has 10 rows and Table B has 10 rows, the result is 100 rows. It is rarely used intentionally in production, but often happens accidentally when a JOIN condition is missed.

## Advanced Reporting (Time Series & Trends)

**Q12. Calculate a running total (cumulative sum) of daily sales.** Use SUM() as a window function ordered by date.

```
SELECT sale_date, daily_revenue,
       SUM(daily_revenue) OVER(ORDER BY sale_date ASC) as cumulative_revenue
FROM daily_sales;
```

**Q13. Calculate the Month-over-Month (MoM) revenue growth percentage.** Use the LAG() function to look at the previous month's revenue in the current row.

```
WITH MonthlySales AS (
  SELECT month, SUM(revenue) as total_revenue FROM sales GROUP BY month
)
SELECT month, total_revenue,
       (total_revenue - LAG(total_revenue) OVER(ORDER BY month)) /
       LAG(total_revenue) OVER(ORDER BY month) * 100 AS mom_growth_pct
FROM MonthlySales;
```

**Q14. How do you pivot rows into columns (e.g., total sales by status)?** Use SUM() combined with a CASE WHEN statement (often called conditional aggregation).

```
SELECT
  user_id,
  SUM(CASE WHEN status = 'Completed' THEN amount ELSE 0 END) as completed_sales,
  SUM(CASE WHEN status = 'Refunded' THEN amount ELSE 0 END) as refunded_sales
FROM transactions
GROUP BY user_id;
```

**Q15. Write a query to get the last 5 records inserted into a table.** Sort descending by the primary key or timestamp, limit to 5, and wrap it in a subquery to re-sort ascending if needed.

```
SELECT * FROM (
  SELECT * FROM audit_logs ORDER BY created_at DESC LIMIT 5
) subquery ORDER BY created_at ASC;
```

## Edge Cases & Data Engineering Concepts

**Q16. Find the missing sequence numbers (gaps) in a table of invoices.** Compare the current ID to the next ID using LEAD( ). If the difference is greater than 1, there is a gap.

```
SELECT invoice_id + 1 AS gap_start,
       next_invoice_id - 1 AS gap_end
FROM (
  SELECT invoice_id, LEAD(invoice_id) OVER(ORDER BY invoice_id) AS next_invoice_id
  FROM invoices
) t
WHERE next_invoice_id - invoice_id > 1;
```

**Q17. What is the difference between DELETE and TRUNCATE?**

- DELETE is a DML command. It removes rows one by one, logs every deletion, and can be rolled back.
- TRUNCATE is a DDL command. It instantly drops the data pages, is much faster, and generally cannot be rolled back.

**Q18. How do you optimize a query that is running too slow?** (Standard DE answer): "I check the EXPLAIN plan to look for Full Table Scans. I ensure JOIN and WHERE columns are indexed. I remove SELECT \* to reduce I/O, and I rewrite subqueries as CTEs or adjust non-SARGable filters."

**Q19. Find users who logged in on 3 consecutive days.** Use the LEAD( ) function to look ahead 1 and 2 days for the same user.

```
WITH NextLogins AS (
  SELECT user_id, login_date,
         LEAD(login_date, 1) OVER(PARTITION BY user_id ORDER BY login_date) as next_day,
         LEAD(login_date, 2) OVER(PARTITION BY user_id ORDER BY login_date) as
day_after_next
  FROM user_logins
)
SELECT DISTINCT user_id
FROM NextLogins
WHERE next_day = login_date + INTERVAL '1 day'
AND day_after_next = login_date + INTERVAL '2 days';
```



**Q20. What is an "Upsert"?** An Upsert (often written as MERGE or INSERT . . . ON CONFLICT) is a command that tries to INSERT a new record, but if a primary key conflict occurs, it will UPDATE the existing record instead. It is the backbone of idempotency in data pipelines.

**Quick Tip:** In a live interview, always talk through your logic *before* you start writing the SQL. Interviewers care more about your problem-solving approach than whether you forgot a semicolon!

## 10. Pro Tips & Best Practices: Writing Production-Grade SQL

Anyone can write a query that runs. A Senior Data Engineer writes queries that are easy to read, easy to troubleshoot, and easy for the next engineer to maintain. Code is read ten times more often than it is written.

### 10.1 Naming Conventions (The "Self-Documenting" Database)

In a production data warehouse, naming conventions are non-negotiable. Bad names lead to bad assumptions, which lead to broken pipelines and incorrect dashboards.

- **Use snake\_case:** Always use lowercase letters separated by underscores. Never use spaces or CamelCase in SQL.
- **Be Descriptive, Not Cryptic:** Avoid abbreviations unless they are universally understood (like `id` or `qty`).
- **Boolean Flags:** Prefix true/false columns with `is_`, `has_`, or `can_`.
- **Meaningful Aliases:** Never use `a`, `b`, and `c` as table aliases. Use the first letter or an acronym of the table name.

#### Example 1: Bad vs. Good Column Names

```
-- ❌ BAD: Cryptic and inconsistent
SELECT
  ActFlg,
  dt,
  usrNme
FROM Users;

-- ✅ GOOD: Clear, predictable, and self-documenting
SELECT
  is_active,
  created_date,
  username
FROM users;
```

#### Example 2: Bad vs. Good Table Aliasing

```
-- ❌ BAD: If this query gets to 100 lines, you will forget what 'a' and 'b' are.
SELECT a.id, b.amount
FROM customers a
JOIN orders b ON a.id = b.cust_id;
```

```
-- ✓ GOOD: Aliases immediately tell you which table the column comes from.
SELECT cust.customer_id, ord.order_amount
FROM customers cust
JOIN orders ord ON cust.customer_id = ord.customer_id;
```

## 10.2 Query Readability & Formatting

SQL is a highly flexible language—it doesn't care about spaces or line breaks. However, your human coworkers care deeply. Adopt a strict formatting style and stick to it.

- **Capitalization:** Write reserved SQL keywords (SELECT, FROM, WHERE, JOIN) in **UPPERCASE**, and all table/column names in **lowercase**.
- **Indentation:** Right-align or clearly indent your core keywords. Put each column in the SELECT clause on a new line.
- **The AND Stack:** Align your AND / OR conditions in the WHERE clause so they form a neat column.
- **Comments:** Explain the *why*, not the *what*. We can see *what* the code does; tell us the business logic behind it.

### Example 1: The "Spaghetti Code" (✗ BAD)

```
select id,name,status from users join orders on users.id=orders.user_id where status='active'
and orders.amount>100;
```

### Example 2: The "Data Engineer" Standard (✓ GOOD)

```
-- Exclude test accounts and find high-value active users for the Q3 marketing campaign
SELECT
  u.user_id,
  u.full_name,
  u.account_status,
  o.order_amount
FROM users u
INNER JOIN orders o
  ON u.user_id = o.user_id
WHERE u.account_status = 'active'
  AND o.order_amount > 100
  AND u.email NOT LIKE '%@smartdatacamp.com'; -- Filtering out internal employees
```

## 10.3 Debugging Tips for Complex Pipelines

When a 500-line SQL script throws an error or returns the wrong numbers, you need a systematic way to find the bug without losing your mind.

**Debugging Tip 1: The "Divide and Conquer" CTE Method** If you have a query with 5 chained CTEs, don't run the whole thing at once. Change the final SELECT statement to pull from the first CTE. If the data looks right, change it to pull from the second CTE. Repeat until you find the step where the data breaks.

```
WITH step_one AS (...),
     step_two AS (...),
     step_three AS (...)
-- Change this final line to debug step-by-step
-- SELECT * FROM step_one LIMIT 100;
SELECT * FROM step_three;
```

**Debugging Tip 2: Commenting out the WHERE clause** If a query returns zero rows, the problem is almost always in the WHERE or JOIN conditions. Comment out the WHERE clause lines one by one using -- to see which condition is aggressively filtering out all your data.

```
SELECT user_id, email
FROM users
WHERE is_active = true
-- AND signup_date > CURRENT_DATE <-- Ah! Found the bug. This was filtering everything
out!
AND region = 'North America';
```

**Debugging Tip 3: The GROUP BY Duplication Test** If your final revenue number is way too high, you probably created a Cartesian product (accidental duplicate rows) via a bad JOIN. To find the duplicates, group by your primary key.

```
-- If this query returns any rows, you have duplicates in your pipeline!
SELECT transaction_id, COUNT(*)
FROM final_reporting_table
GROUP BY transaction_id
HAVING COUNT(*) > 1;
```

**Quick Tip: Embrace version control.** Never write a massive SQL script directly in your database UI (like Snowflake or DBeaver) without saving it. Write your queries



in an IDE (like VS Code) and commit them to Git. If you break a working query, you can easily revert to the previous version.

---

## **Congratulations on Completing the SQL Architect's Cheat Sheet!**

You now have the syntactical foundation and the senior-level patterns required to write production-grade data pipelines. But reading about SQL is only the first step. To truly master Data Engineering, you need to build.

**Want to master Data Engineering, Apache Spark, and SQL?** Visit **[SmartDataCamp.com](https://www.smartdatacamp.com)** and enroll in our project-based courses. We guide you step-by-step through building real-world, cloud-based data pipelines that will make your resume stand out to top tech companies. Start your journey today!